

PREVOS-DZ: A Short-Mid Term Algerian Electric Load Forecasting Software

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Abstract— Electricity demand is nowadays a very important development factor for growing economies. In order to support daily electrical demand and meet the customer daily needs, electricity and companies operators must rely on powerful and precise forecasting tools. PREVEOS-DZ was designed and implemented in order to predict Short and Midterm electrical energy and power demand in order to plan day to day production planning as well as yearly electrical development established by the Algerian Operator System. The software is based, on the multiple regression approach as well as Artificial Neural Networks, especially for power forecasting, offering models in both paradigms. The Forecast strategy is implemented not only for global national forecast but also for regional demands, even if the electrical grid is fully connected. PREVOS-DZ is implemented using C # and allows full compatibility with Excel tabular, used mostly by the System Operator. The results of PREVELOS-DZ obtained as a prevision exercise were validated on data from 2018 and presented in this paper.

Keywords—Short-Mid-term load forecasting, Multiple linear regressions, Artificial Neural networks.

I. INTRODUCTION

Electrical grids are complex interconnected growing systems. Their size is proportional to the country or area development as well as customer's numbers and economic needs, making them very hard to manage from the generation and transmission of energy to distribution circuits. This paper deals with the management of the production side, a complex problem as electricity is not "yet" storable at large scale. For this reason, producers need to predict load consumption to ensure good load balancing between supply and demand instantaneously.

Different forecast scenarios are then suitable for different management situations; one may cite three scenarios based on forecast horizons: short term (from 1 h to 1 week ahead), medium term (a week to several months or a few years) and long term (more than 1 year in advance). Respectively, short, medium and long terms forecast are suitable for: hour to hour electricity production management planning, production unit's management and maintenance and design and construction of future production units to fulfill future needs.

This paper deals with Short-term electric load forecasting (STLF). STLF involves scheduling of generators capacity to minimize daily running costs, affecting transmission planning

by dispatching the produced power between customers and system operations in real time. This is to ensure that the system operates in a reliable and secure manner. In order to build reliable STLF models, many paradigms and approaches have been proposed in the literature. One may divide the approaches into two main categories:

On the one hand, traditional approaches including regression model classical time-series models [1] or Autoregressive Moving Average (ARMA) models as in [2]. Sargunraj et al. [3] proposed a two stage approach based on Kalman filtering [4]. Here the mean load is predicted consequently Kalman filter is applied in real time using an error feedback from previous hours. Kyriakides and Polycarpou [5], discusses both approaches and draw conclusions.

On the other hand, Artificial Intelligence (AI) and machine learning based techniques proved very efficient lately with growing available computational power. Artificial Neural Networks (ANNs) [6] have seen many successful experiments and practical applications usage [7]. Other AI paradigms such as, Support Vector Machine (SVM) [8] have been also used. In [9] SVMs interpolate between load and temperature data. Other tried fuzzy logic techniques to manage abnormal situations in load demand [10].

Various other hybrid approaches of artificial intelligence techniques have also been applied to increase model efficiency [11–13]. More recently, Lusi et al., in [14], investigated how calendar effects, granularity and training set length affect the accuracy of a day-ahead load forecast for residential customers. Lately deep learning architectures permits the inclusion of more input influential data extracting the best features at first then performing forecasting [15]. This approach cannot be used at present for the Algerian market due to insufficient gathered data.

The paper is organized as follows: a non-exhaustive description of both used approaches for instance: Multiple Linear regression (MLR) and Artificial Neural Networks (ANN) are presented in Section 2 and 3. In the fourth section, the framework of PREVOS-DZ is presented with experiments

and results shown in Section five. Finally, a conclusion on both technical and cooperation aspects is drawn.

II. MULTIPLE LINEAR REGRESSION

A. Principles

Regression analysis is a linear approach consisting in finding the relationship between a set of input output data, determining causality between those. In the case of Electrical Load, the causality has to be determined between exogenous inputs such as: temperature, humidity, etc. past electrical load and future predicted load. Finding a suitable model rely solemnly on valid informative data collected from the underlined variables of interest. Regression is then used to estimate the quantitative effect of the causal variables upon the variable that they influence. Regression techniques have long played a central role in the field of economic statistics ("Econometrics").

Multiple Linear regressions is used to establish PREVOS-DZ models, examining the relationship between a set of independent variables on a single dependent one. A multiple linear regression model with more than one explanatory variable can be given in equation (1):

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p + \varepsilon \quad (1)$$

Where y is the output variable, β_i are regression parameters ($i = 1, 2, \dots, p$), x_i are the input variables ($i = 1, 2, \dots, p$) and ε is the random error term.

Exponential Regression is of crucial importance when the tendency of a temporal series is of exponential nature. This is the case of load growth in developing countries and more precisely in Algeria. Exponential regression may also be used producing an exponential curve that best fits a set of data that does not change linearly with time [16]. An exponential regression model is mainly used for long term predictions with the presence of a trend, and is written as follows:

$$y = \beta_0 \times \beta_1^{x_1} \times \beta_2^{x_2} \times \dots \times \beta_p^{x_p} \quad (2)$$

Where β_i are parameters for the n independent variables.

B. MLR models for Load forecasting

In MLR (Multi Linear Regression) models, inputs are composed of: lagged load values based on autocorrelation analysis as well as nonlinearly related exogenous inputs, in this case hourly temperature estimations.

Electric load L at hour t is therefore explained by several selected past values, in addition to temperature estimations: L_{t-1} , L_{t-2} , L_{t-3} , L_{t-24} and L_{t-168} representing, respectively, the three previous hourly loads, the last day load, and last week load at the same time of the day. Furthermore, the context of day type is represented using one hot encoding for each day of the week (DoW) as well as an additional input for special days (ToD), resuming to a total of 11 input variables as described in equation (3).

$$L_t = a_1 L_{t-1} + a_2 L_{t-2} + a_3 L_{t-3} + a_4 L_{t-24} + a_5 L_{t-168} + b_1 T_{t-1} + b_2 T_{t-2} + b_3 T_{t-3} + b_4 T_{t-24} + DoW + ToD \quad (3)$$

For a sampling of 15mn, the above model equations will contain indices multiplied by 4 (i.e., 1h = 4 times 15mn).

III. ARTIFICIAL NEURAL NETWORKS

Artificial Neural Network (ANN) is a data processing approach inspired by biological nervous systems, which underlines the ways in which the brain processes information. McCulloch and Pitts [17] proposed the first mathematical and computational model of biological neurons in 1943. They were the first to show the formal functions.

The characteristics of ANN information processing make it a powerful computational technique which is able to learn from examples, and then, generalize to others cases that were not presented during training. ANNs are composed of a set of neurons where each neuron is autonomous and undergoes two phases:

The computation of the weighted sum of the inputs in is given by the following equation:

$$a = \sum_i w_i \times e_i \quad (4)$$

where w_i and e_i are respectively the weight and the input of the neuron i . A transfer function (or activation function) calculating the value of the state of the neuron, based on the previously computed sum.

The major advantages of ANN's derive from the fact that connections between neurons are scalable and are the result of a learning process. Once fixed, the ANN may possess the ability to solve any non-linear problems and therefore approximate any nonlinear function.

For that matter, and based on connections between neurons, several topologies have been proposed in the literature, such as Feed Forward [18], Recurrent Neural Network [19], Self Organizing Map (SOM) [20], etc. In this study, Non-Linear Auto Regressive Neural Network with Exogenous Inputs (NARX) Multi Layer Perceptron (MLP) is used (see Fig. 1). The model consists of a recurrent neural network inspired by the linear Auto Regressive with Exogenous Inputs (ARX) model. The NARX model is composed by an input, a hidden layer and a one neuron output layer, like the Feed Forward neural network, with the exception that there is a feedback from the output layer to the input layer (shown in Fig. 1). The output is given by the following equation:

$$\hat{y}_t = \sum_{i=1}^d w_i \cdot y_{t-1} + \sum_{j=1}^m w_j \cdot T_{t-1} + w_{DoW} \cdot DoW + w_{ToD} \cdot ToD \quad (5)$$

where w are the weights to be estimated, d is the autoregressive delay and m the exogenous input delay on the exogenous variable T being here the temperature, and DoW and ToD the two hot encoded inputs described earlier.

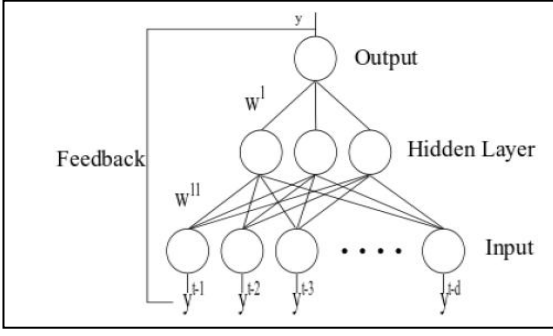


Fig. 1. Feed Forward MPC Neural Network.

In ANN, instead of using a standard linear regression, the parameters are estimated using a learning algorithm e.g. the Retro propagation algorithm, the Levenberg Marquardt algorithm, or resilient propagation, etc. The added feature of the recurrent model, is the inclusion of past model outputs within the input vector. This ensures multi-step ahead predictions and allows the use of more nonlinear model types such as Nonlinear Box Jenkins.

IV. AVAILABLE DATA

A. Exogenous Inputs (Temperature)

For the purpose of this work, daily hourly temperature load are estimated based on minimum T_{min} and maximum T_{max} daily temperatures provided by SONEGGAZ.

Constructing temperature daily load is based on idealized temperature curve according to Linvill equations [21], in turn based on the assumption that the time of maximum daily temperature is 2 h after solar noon and the temperature cycle after sunrise follows a sine curve:

$$T_{ih} = (T_{maxi} - T_{mini}) \sin\left(\frac{\pi t_a}{DL_i - 4}\right) T_{mini} \quad (6)$$

where T_{ih} is temperature at hour h and T_{maxi} and T_{mini} are the maximum and minimum temperature at day i . DL_i is the number of hours from sunrise to sunset, and t_a is the time in hours (≥ 0) counted from sunrise.

After sunset, the temperature is assumed to follow a logarithmic decline:

$$T_{ih} = T_s - \frac{T_s - T_{mini+1}}{\ln(24 - DL_i)} \ln(t_b) \quad (7)$$

T_s is the sunset temperature obtained after sunrise as determined from Eq. (6), and T_{mini+1} is the minimum temperature of the following day $i+1$, which is replaced by the minimum temperature of the same day for the hours before sunrise. The time t_b indicates the time in hours after sunset plus 1 h ($t_b \geq 1$). Linvill's equations also require some input parameters such as: sunset time, sunrise time and day length, computed applying [22] and [23] procedures based on geographic latitude.

An example of temperatures estimation daily profiles generated using the above method is shown in Fig. 2. Variations between seasons are clearly noticeable when looking at minimum and maximum temperatures. Minimum temperatures always occur at sunrise because of two mechanisms.

First, the Earth surface stops heating up as solar radiation is not being absorbed after sunset. Secondly, conduction and convection heat energy transport up into the atmosphere is operating. Warm air that was at the surface is then replaced by cooler air from above due to atmospheric mixing [24].

The obtained results prove a better approximation of temperature daily load when compared to linear and cubic interpolations. The increase and decrease of temperature when minimum and maximum temperatures are reached. Moreover, estimation of temperature profile helps the model to achieve better performances as exogenous inputs are better represented [25].

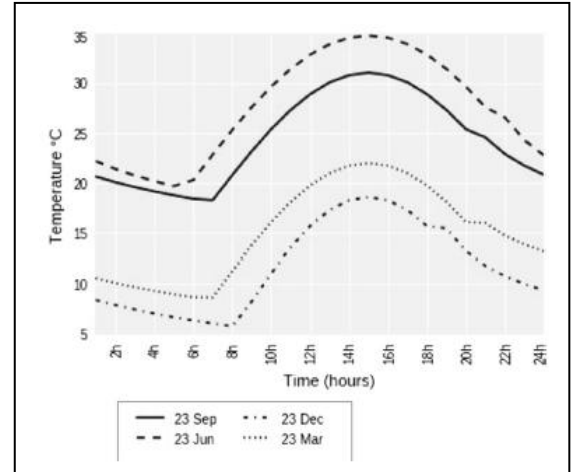


Fig. 2. Temperature profile estimation results

B. Load Data

The load data used, represents the Algerian quarterly and hourly electric load for a 4-year period (2013–2017). Prior logged data are omitted as they will badly influence model learning and optimal weight and biases search. Indeed, the data showed important behavioral load changes from 2013 and beyond with the widespread of air conditioners. Therefore data after 2013 are used as they better represent actual load shape.

In order to better validate and evaluate induced enhancement in performance given by ANN, see section III, a traditional technique of time-series analysis in the form of a multiple linear regression, see section II is used as a benchmark.

Data are then separated into three subsets: the training set (60%) for the creation of the model, validation set (20%) for the optimization of hyper-parameters to avoid over fitting and the last data set for testing (20%),

A. Développé models

Not less than 216 models were developed for the purpose of the PREVOS-DZ tool. Models have been validated jointly by the authors from both institutions (academic and industrial). Models cover a large prediction needs, classified and chosen according to:

- Model type (day type, special periods and days, etc.)
- Region Models (National or Regional)
- Saison and sometimes monthly models
- Prediction Horizons (quarterly, hourly and weekly models)

Table 1 and 2 give an overview of the number of developed models classified by types, for respectively MLR and ANN models.

TABLE I. " DEVELOPED MLR MODELS

Type	National		Annaba		Sétif	Oran	Alger		Sud		Adrar	
	15m	1 H	15m	1 H	1H	1H	15m	1 H	15m	1 H	15m	1 H
Days (Sat, WD, WE)	3	3	3	3	3	3	3	3	3	3	3	3
Season	4	4	4	4	4	4	4	4	2	2	2	2
Month	12	0	0	0	0	0	0	0	0	0	0	0
Ramadan	1	1	1	1	1	1	1	1	1	1	1	1
Global	1	1	1	1	1	1	1	1	1	1	1	1
Total/Region	21	9	9	9	9	9	9	9	7	7	7	7
Total RLM	112											

TABLE II. " DEVELOPED ANN MODELS

Type	National		Annaba		Sétif	Oran	Alger		Sud		Adrar	
	15m	1 H	15m	1 H	1H	1H	15m	1 H	15m	1 H	15m	1 H
Days (Sat, WD, WE)	3	3	3	3	3	3	3	3	3	3	3	3
Season	4	4	4	4	4	4	4	4	2	2	0	0
Month	12	0	0	0	0	0	0	0	0	0	0	0
Ramadan	1	0	1	1	1	1	1	1	1	1	0	0
Global	1	1	1	1	1	1	1	1	1	1	0	0
Total/Region	21	9	9	9	9	9	9	9	7	7	3	3
Total ANN	104											

B. Models accuracy

In order to assess model quality, Mean Absolute Error (MAE) and Mean Absolute Percentage Error (MAPE) [22] are used to obtain comprehensive and objective evaluation for the performance of the proposed approach.

Overall results in terms of MAPE, are presented in Table 3 and graphical evaluation for a 24hour forecast is given for random days in Figures 3, 4 and 5 for different model types. Fig. 3, shows the prediction using a quarterly overall national model for a summer day (2nd June 2016). It can be seen that the daily seasonality is well respected and that the ANN model performs

better than the MLR one. The case is even more strengthened for the Ramadan national model, Fig. 4, where the ANN model is even more accurate when compared to the MLR one.

Fig. 5, shows an example of regional global models, focusing here on the region of Annaba. It can be seen that regional load is less regular than national one, and thus harder to fit. This is the case for all regions for, as external factors affect more heavily regional load due to their lower amplitude, but not only. Nonetheless, the ANN model still outperforms the MLR one. Although it is the case for most model types, and for the average evaluation, for some specific days the MLR model might outperform the ANN one.

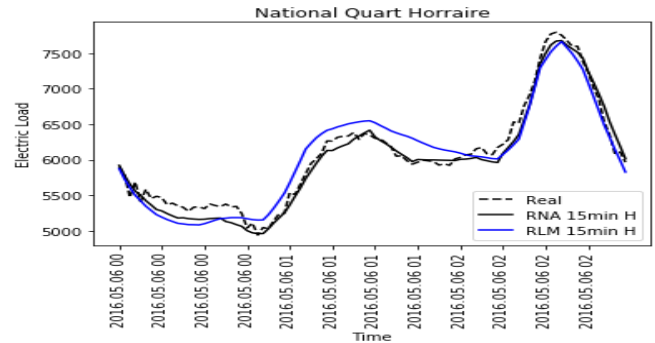


Fig. 3. National global model performances for (2nd June 2016)

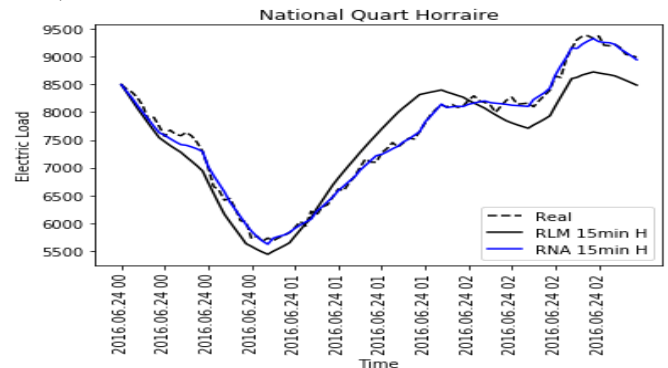


Fig. 4. National Ramadan model performances (24th June 2016)

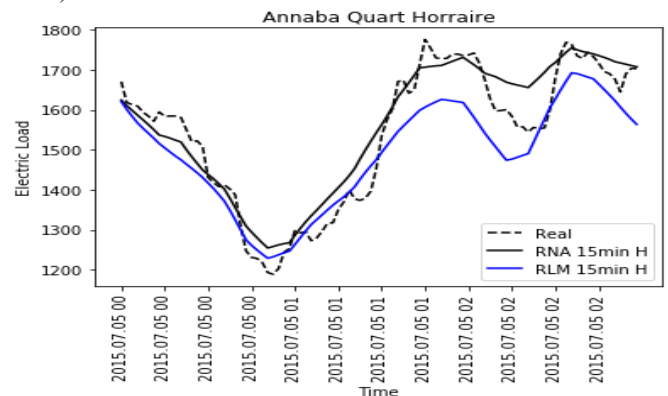


Fig. 5. Regional (Annaba) Global model performances (5th July 2016)

TABLE III. ⁿ PERFORMANCES OF NATIONAL DEVELOPED MODELS

MLR	1 Hour ahead			24 Hours ahead		
	MAE (MWh)	MAPE	%	MAE (MWh)	MAPE	%
Global Model (1h)	173.38	2.19		404.07	5.25	
Seasonal Models (1h)	127.27	1.61		202.64	2.92	
Day Type Models (1h)	158.63	1.97		383.99	4.98	
Monthly Models (1h)	98.97	1.27		198.47	2.54	
Ramadan Model (1h)	93.81	1.12		232.36	2.75	
Global model (15mn)	319.06	4.10		395.29	5.11	
Seasonal Models (15mn)	318.15	4.08%		390.06	5.03%	
Day Types Models (15mn)	318.08	4.08%		368.33	4.73%	
Ramadan Model (15mn)	96.58	1.18%		234.40	2.84%	
ANN	1 Hour ahead			24 Hours ahead		
	MAE (MWh)	MAPE	%	MAE (MWh)	MAPE	%
Global Model (1h)	126.18	1.58%		466.74	6.52%	
Seasonal Models (1h)	118.08	1.52%		241.10	3.14%	
Day Type Models (1h)	126.20	1.62%		210.39	2.61%	
Monthly Models (1h)	122.18	1.58%		203.52	2.59%	
Ramadan Model (1h)	122.81	1.49%		273.92	3.21%	
Global model (15mn)	109.16	1.43%		424.77	6.36%	
Seasonal Models (15mn)	116.63	1.53%		288.01	3.69%	
Day Types Models (15mn)	136.59	1.78%		472.76	6.88%	

From Table III, one may draw some occlusions:

- ⁿ Obviously 1 hour ahead models are more accurate than 24hours one with a maximum MAPE error of 1.78% for ANN models and 4.08% for MLR ones.
- ⁿ Overall, ANN models perform better than MLR, except for the case of 24 hours ahead using a sampling time of 1H.
- ⁿ Seasonal and day type models outperform global ones. This is logical as those models are trained with just the specific data of the model type or season, thus focusing only on the load profile of those. This makes every day type/seasonal model very good within its specific data range.
- ⁿ Using an estimated temperature daily profile instead of two values (min and max) enhance the prediction results, see [23].

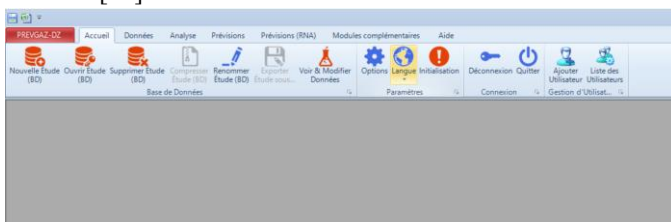


Fig. 6. PREVOS-DZ main window

VI.ⁿ PREVOS-DZ GRAPHICAL USER INTERFACE

All the developed models Tables I and II, as well as other functionalities and data are regrouped to constitute the prevision tool PREVOS-DZ. Fig. 6 shows the main windows containing the link to all 6 remaining tabs. The main window contains the log in button as well as the access to existing saved studies, and the option of creating new ones.

PREVOS-DZ, contains the following Tabs:

- ⁿ **Data:** Access and addition to all data (Electricity load for all regions as well as exogenous inputs, mainly temperature data), Fig. 7.
- ⁿ **Profiles:** This Tab contains all the necessary tools to create load profiles from existing data, as well as manually.
- ⁿ **MLR Prediction and ANN Prediction:** These Tabs allow the user to make any forecast based on all the already developed models. Fig. 8 shows a prediction for the national case based on seasonal models for the 19th of March 2018.
- ⁿ **Weekly predictions:** This Tab permits the prediction of weekly forecast up to a year ahead. Te user may also obtain hourly prediction based on the weekly predicted peak and the associated profile, Fig. 9.
- ⁿ **Real time:** This Tab implements previous models using a real time approach. A 24 hour ahead quarterly predictions are computed each 15mn as soon as the system accesses real logged load. The predictions are then automatically updated using real data.
- ⁿ **Power to energy conversion:** All daily forecasts can be converted to energy using two different approaches based on the sum of obtained hourly powers or calculation of the integral surface of the daily forecast.

VII.ⁿ CONCLUSIONS

Artificial Neural Networks and more precisely, the multi Layer Perceptron, are powerful tools for the modeling and prediction of time series, proven here with the prediction of Algerian electrical Load. ANNs were proven more efficient than classical MLR models for most cases. However, some special days were not accurately predicted, with MAPE errors above 9%. These days, are considered special days with a load profiles far from the average load of the period, day type or region.

This is mainly due to external exogenous factors, called physical factors such as: drill fires causing abnormal temperature rise, important non periodic events such as soccer games, political events, inducing important habit changes at certain times, severe meteorological conditions, etc.

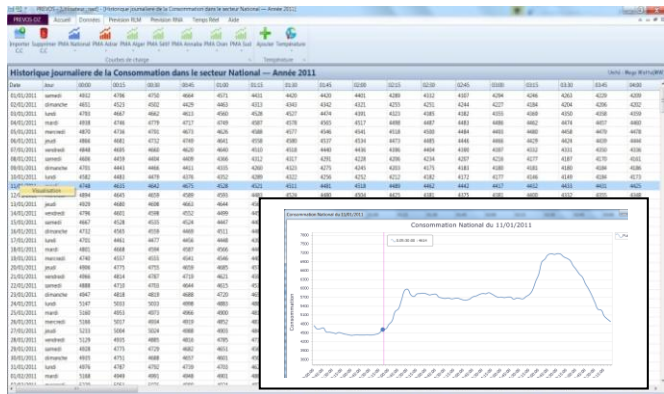


Fig. 7. Data Tab, with graphical options for data visualization

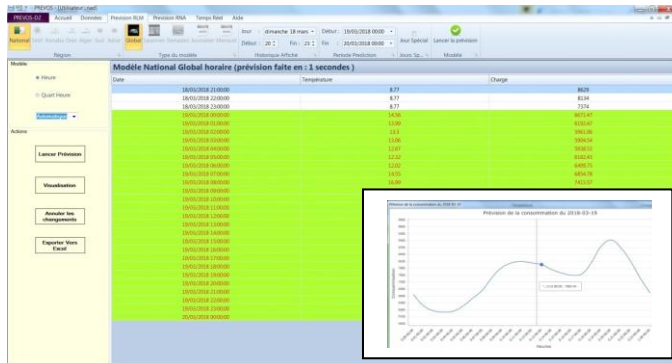


Fig. 8. MLR prediction using the national seasonal models

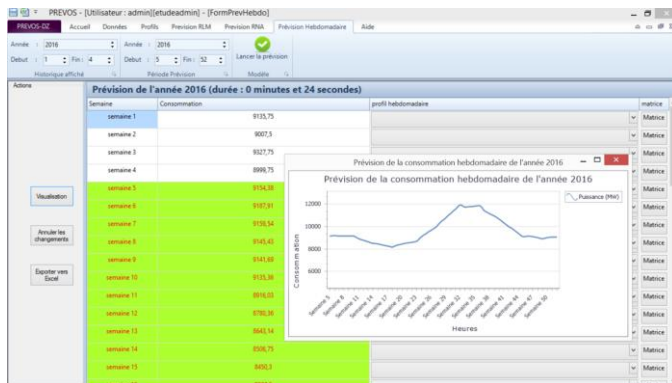


Fig. 9. Weekly predictions based on MLR models

Finally, offering a complete tool, including not only valid prediction models for every region, season and day type but also a midterm weekly prediction models and other profile creation and MLR model creating tools, is a valuable asset for our partner OS (Opérateur Système). Indeed, the developed tool, will certainly enhance the predictions made using previous tools and thus, offers better decision making. Furthermore, a strong partnership with a university laboratory, close to the OS, will permit to further develop the tool introducing new solutions for yet unsolved problems.

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